

POLYNOMIAL POTENTIALS AND A HIDDEN SYMMETRY OF THE HILL-DETERMINANT EIGENVALUE METHOD

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For the potential $V(x) = g_1 x^2 + g_2 x^4 + g_3 x^6$ and its polynomial generalizations, the "true" and "false" energies of Killingbeck may both be interpreted as physical, corresponding to the two different forms of the potential. Thus, the Hill-determinant method has two non-equivalent forms, and neither of them gives spurious solutions.

During the last few years, the Schrödinger eigenvalue problem

$$(-d^2/dx^2 + g_1 x^2 + g_2 x^4 + g_3 x^6)\psi(x) = E\psi(x),$$

$$g_3 > 0, \quad \|\psi\| < \infty, \quad (1)$$

has attracted the attention of a number of authors [1-4]. They employed the simple power-series ansatz

$$\psi(x) = \exp(-\frac{1}{4}\alpha x^4 + \frac{1}{2}\beta x^2) \sum_{n=0}^{\infty} A_n x^{2n+k},$$

$$k = 0, 1, \quad \alpha = g_3^{1/2} > 0, \quad (2)$$

and converted the differential equation (1) into the recurrences

$$b_n A_{n+1} + a_n A_n + c_n A_{n-1} + d_n A_{n-2} = 0,$$

$$b_n = (2n + k + 2)(2n + k + 1),$$

$$a_n = (4n + 2k + 1)\beta + E,$$

$$c_n = \beta^2 - (4n + 2k - 1)\alpha - g_1,$$

$$d_n = d_2 = 2\alpha(\beta_0 - \beta),$$

$$\beta_0 = -g_2/2\alpha, \quad A_{-1} = A_{-2} = 0, \quad n = 0, 1, \dots \quad (3)$$

Tentatively, they have chosen $\beta = \beta_0$ (i.e., $d_n = 0$, $n = 2, 3, \dots$) and postulated the so-called Hill-determinant condition

$$\det Q_N = 0,$$

$$Q_N = \begin{pmatrix} a_0 & b_0 & 0 & \dots & & \\ c_1 & a_1 & b_1 & 0 & \dots & \\ \vdots & & & & & \\ 0 & \dots & & 0 & d_N & c_N & a_N \end{pmatrix}, \quad (4)$$

as a quasivariational secular equation. For $g_2 > 0$, such a postulate proved to be correct – the $N \gg 1$ roots $E_n^{(N)}$ of eq. (4) have been shown [3] to reproduce the physical energies $E_n^{(\text{true})}$ in the infinite-dimensional limit,

$$E_n^{(\text{true})} = \lim_{N \rightarrow \infty} E_n^{(N)}, \quad g_2 > 0, \quad \beta = \beta_0,$$

$$n = 0, 1, \dots \quad (5)$$

For $g_2 \leq 0$, the physical energies may be obtained from eq. (4) only after a thorough and rather complicated analytic-continuation [4] or fixed-point-expansion [5] modification of Q_N . An alternative removal of the $g_2 \geq 0$ asymmetry proved to lie simply in a removal of the artificial restriction $\beta = \beta_0$ [6,7].

In ref. [6], an analysis and interpretation of the $N \gg 1$ form of eq. (4) as an asymptotic boundary condition pertinent to the difference Schrödinger equation (3) has been used. As a consequence, the

improved Hill-determinant prescription

$$E_n^{(\text{true})} = \lim_{N \rightarrow \infty} E_n^{(N)}, \quad \beta < \beta^{(\text{cnt})} = g_2/2\alpha, \\ n = 0, 1, \dots, \quad (6)$$

acquired a rigorous mathematical foundation. It determines the spectrum of binding energies for all the couplings in the sextic anharmonic oscillator potential (1). Its asymmetry with respect to the auxiliary parameter β seems natural.

By means of a purely numerical analysis of eq. (4) in a "forbidden" domain of parameters $\beta > \beta^{(\text{cnt})}$, an interesting observation has been made by Killingbeck [7] – a "false" $N \rightarrow \infty$ limit

$$E_n^{(\text{false})} = \lim_{N \rightarrow \infty} E_n^{(N)}, \quad \beta > \beta_1, \quad n = 0, 1, \dots, \quad (7)$$

seems to exist and exhibits a number of symmetry properties similar to the physical case with $E_n^{(\text{true})}$. In the present letter, we shall describe these symmetries in an explicit way.

In the first step of our considerations, we notice that the matrix elements $a_n = a_n(\beta, E) = -a_n(-\beta, -E)$ and $d_n = d_n(\beta, g_2) = -d_n(-\beta, -g_2)$ are all asymmetric functions of the parameters β , g_2 and E . At the same time, the remaining matrix elements remain independent of this sign-change, so that

$$Q_N = \begin{pmatrix} 1 & & & & \\ & -1 & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & (-1)^N \end{pmatrix} \\ \times \tilde{Q}_N \begin{pmatrix} -1 & & & & \\ & 1 & & & \\ & & -1 & & \\ & & & \ddots & \\ & & & & (-1)^{N+1} \end{pmatrix} \quad (8)$$

where $Q_N = Q_N(\beta, E, g_1, g_2, g_3)$ and $\tilde{Q}_N = Q_N(-\beta, -E, g_1, -g_2, g_3)$. As a consequence, we have (cf. ref. [5] for $\beta = \beta_0$)

$$\det Q_N(\beta, E, g_1, g_2, g_3) \\ = (-1)^{N+1} \det Q_N(-\beta, -E, g_1, -g_2, g_3), \quad (9)$$

so that the roots $E_n^{(N)} = E_n^{(N)}(\beta, g_1, g_2, g_3)$ of eq. (4) exhibit the following symmetry

$$E_n^{(N)}(\beta, g_1, g_2, g_3) \\ = -E_n^{(N)}(-\beta, g_1, -g_2, g_3), \quad (10)$$

In the next step, let us choose, e.g., the "physical" value of the right-hand-side parameter $-\beta < -g_2/2\alpha$. With respect to the results of ref. [6], we get $-E_n^{(\text{true})}$ here in the $N \rightarrow \infty$ limit. Of course, the sequence of the left-hand-side limits decreases with increasing n and must be unphysical. Since $\beta > g_2/2\alpha$, we may identify $\beta_1 = \beta^{(\text{cnt})}$ in (7) and get

$$E_n^{(\text{false})}(g_1, g_2, g_3) = -E_n^{(\text{true})}(g_1, -g_2, g_3). \quad (11)$$

This is precisely the functional relationship expected by Killingbeck and recovered by himself numerically for $g_2 = 0$.

Our results (8)–(11) admit a straightforward generalization. Indeed, whenever we replace eq. (1) by the polynomial problem

$$\left(-\frac{d^2}{dx^2} + \sum_{n=1}^{2q+1} g_n x^{2n} \right) \psi(x) = E \psi(x), \\ g_{2q+1} = \alpha_q^2 > 0, \quad \|\psi\| < \infty, \quad (12)$$

and transform it into a generalized relation (3)

$$Q_\infty \begin{pmatrix} A_0 \\ A_1 \\ \vdots \end{pmatrix} = 0 \quad (13)$$

by means of an ansatz similar to (2),

$$\psi(x) = e^{f(x)} \sum_{n=0}^{\infty} A_n x^{2n+k}, \quad k = 0, 1, \\ f(x) = \sum_{j=0}^q \alpha_j \frac{x^{2j+2}}{2j+2}, \quad \alpha_q = -|\alpha_q| < 0, \quad (14)$$

we may try again to satisfy eq. (8) with the $2q+2$ diagonal matrices $Q_N = Q_N(\alpha_0, \alpha_1, \dots, \alpha_q, E, g_1, g_2, \dots, g_{2q})$. The corresponding change of sign in the odd diagonals may immediately be seen to

correspond to a change of parameters

$$(\alpha_0, \alpha_1, \dots, g_{2q}) \rightarrow (-\alpha_0, \alpha_1, -\alpha_2, \dots, (-1)^{q+1} \alpha_q, -E, g_1, -g_2, \dots, -g_{2q}). \quad (15)$$

Obviously, we must restrict our attention to odd $q = 2p + 1$ since the dominant normalization factor α_q should not change sign in our ansatz (14).

The remaining part of our construction is almost trivial now. In place of (10), we obtain the relation

$$\begin{aligned} E_n^{(N)}(\alpha_0, \alpha_1, \dots, \alpha_{q-2}, \beta, g_1, g_2, \dots, g_{2q+1}) \\ = -E_n^{(N)}(-\alpha_0, \alpha_1, \dots, \alpha_{q-2}, -\beta, \\ g_1, -g_2, g_3, \dots, g_{2q+1}), \\ \beta = \alpha_{q-1}, \quad q = 2p + 1. \end{aligned} \quad (16)$$

Then, we recall the condition of applicability of the Hill-determinant method as obtained in ref. [8],

$$\beta < \beta^{(\text{cnt})} = g_{2q}/2 |\alpha_q| = \beta_1, \quad (17)$$

and, in a way paralleling the preceding argumentation, arrive at the final formula

$$\begin{aligned} E_n^{(\text{false})}(g_1, g_2, \dots, g_{2q+1}) \\ = -E_n^{(\text{true})}(g_1, -g_2, g_3, -g_4, \dots, g_{2q+1}), \\ q = 2p + 1. \end{aligned} \quad (18)$$

We may summarize:

(i) For even parameters q , the existence and/or validity of a relation similar to our final result (18) remains an open problem. At the same time, a change of variables [8] $x^2 \rightarrow x_{\text{new}}$ may be used to modify its form slightly. This will not be done here.

(ii) In effect, our results may be summarized also as a “Wick” rotation in the variable $x \rightarrow ix$ ($i^2 = -1$). This is the most concise formulation of the “hidden symmetry” in question. This symmetry is reflected then by the possibility to use both the small and large parameter β . The latter choice

has, obviously, no relation to the intuitive “truncation” interpretation of the Hill determinants, although it still gives the correct binding energies for the “transformed” potentials.

(iii) The recent debate about the “physical” or “unphysical” asymptotics of $\psi(x)$, $x \rightarrow \infty$ [9] is entirely misleading. Indeed, relation (4) [8] characterizes the appearance of an asymptotic node in $\psi(x)$, so that the ansatz (14) described always an unphysical state with energy lying closely above or closely below the exact value. Then, the limiting transition $N \rightarrow \infty$ in eq. (6) corresponds merely to a shift of the wavefunction node to infinity. In the limit $N \rightarrow \infty$, an infinite number of significant digits would be necessary to distinguish between the “physical” and “unphysical” wavefunctions, which, obviously, makes no sense in practice.

Note added. The symmetry $E(\text{false}) = -E(\text{true})$ relates in fact the two non-equivalent forms of the Hill-determinant prescription using $\beta > \beta^{(\text{cnt})}$ and $\beta < \beta^{(\text{cnt})}$, respectively. In this sense, it need not be restricted to the present class of potentials of course. For example, it still holds numerically for the simplest even- q potential $V(x) = x^2 + \lambda x^4$ with $0 \leq \lambda < 0.05$ (J. Killingbeck, private communication). In accord with the same author, a smooth transition to the small and negative couplings $\lambda > -0.05$ is also possible here: The Hill-determinant roots E seem to represent then the real part of the energy of the long-lived quasi-bound states.

References

- [1] V. Singh, S.N. Biswas and K. Datta, Phys. Rev. D 18 (1978) 1901.
- [2] D.P. Datta and S. Mukherjee, J. Phys. A 13 (1980) 3161
- [3] M. Znojil, Phys. Rev. D 26 (1982) 3750
- [4] D. Masson, J. Math. Phys. 24 (1983) 2074.
- [5] M. Tater, submitted to J. Phys. A.
- [6] M. Znojil, submitted to Phys. Rev. D
- [7] J. Killingbeck, Phys. Lett. A 115 (1986) 301
- [8] M. Znojil, submitted to J. Math. Phys.
- [9] G.P. Flessas and G.S. Anagnostatos, J. Phys. A 15 (1982) L537, A. Hautot and M. Nicolas, J. Phys. A 16 (1983) 2953